Page No.
Date / /

ROC Curve → threshold selection.

- For logistic regression

$\sigma \geq 0.5 \rightarrow 1$

$\sigma < 0.5 \rightarrow 0$

Here, threshold is 0.5

- In lot of senarios 0.5 will be better, but other threshold values may give better result.

In email spam problem:

mistake:

A → Not spam
P → Spam

Dangerous.

A → spam
P → non spam.

for this:

threshold → 0.75.

lower → not spam.

higher → spam.

We use ROC for getting better threshold.

Page No.
Date: / /

	Prediction: 1	0
Actual 1	TP	FN
Actual 0	FP	TN

True Positive Rate [Benefit]:

- $TPR = \frac{TP}{TP + FN}$
- Higher TPR is better.
- Intution of TPR:

  How many spam emails are correctly classified out of total actual?

  $$TRP = \frac{TP}{(TP + FN = \text{Actual (1 total)})}$$

False Positive Rate ~~[Benefit]~~ [loss/cost]:

$$FPR = \frac{FP}{FP + TN}$$

Chorn model:

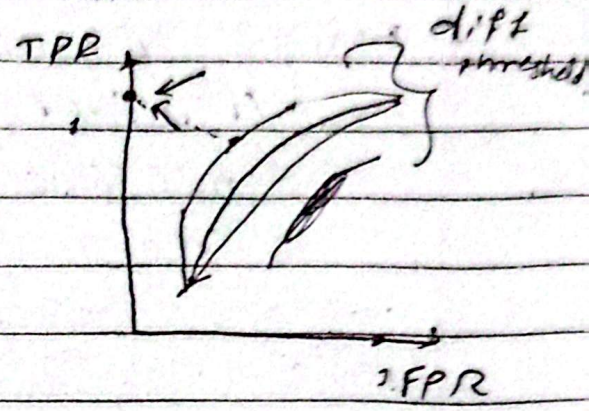
- who won't leave the platform → but, model says they will leave
- not spam email → spam

- Lower FPR is better.

Page No.
Date: / /

ROC Curve is the graph of TPR vs FPR.

- Calculate TPR & FPR for every threshold

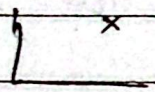


Best: TPR = 1, FPR = 0

Worst: TPR = 0, FPR = 1.

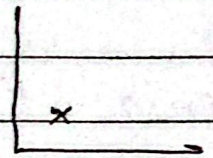
Case a) For low threshold: [around 0]

FPR ↑ & TPR ↑



Case b) For high threshold: [around 1]

- Spam ↓ • TPR ↓
- notspam ↑ • FPR ↓



Finding nearest point near. TPR = 1.

max value of (tpr − fpr)

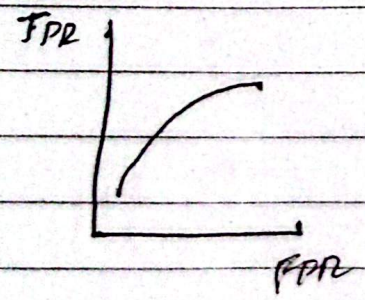
Case c) For threshol (~0.8):

TPR ↑ FPR ↑

But, rate won't be same.

Code

```
from sklearn.metrics import roc_curve
fpr, tpr, thresholds = roc_curve(y_test, y_score)
```




```
optimal_idx = np.argmax(tpr-fpr)
optical_threshold = threshold[optical_idx]
```

argmax → find index of maximum value of tpr−fpr

y_score = probability → clf.predict_proba(pos)
[0.5, 0.2, 0.9]

Page No.
Date: / /

Reason of using ROC
- threshold value
- AUC-ROC for comparing 2 separate model.

AUC - ROC : Area under curve of ROC curve

AUC value higher, higher model performance

~~find~~
Code:

```
from sklearn.metrics import roc_curve,
                            roc_auc_score.
```

find for both models { find fpr, tpr, threshold → roc_curve.
lr_auc = roc_auc_score(y_test, ~~y_pred~~ y_score).

find roc_auc_score.

higher roc_auc_score better